Government Polytechnic Dahod Microproject

Report File on

Smart Street Light System

FIoT(4360703)

Submitted by

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Acknowledgement

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Introduction

A Smart Street Light System is an advanced lighting solution that uses **Internet of Things (IoT)** technology to automatically control street lights based on environmental conditions such as light intensity and motion detection.

Traditional street lights consume a lot of electricity because they remain ON throughout the night. This project aims to reduce energy consumption and improve efficiency by using sensors and automation.

Objectives

1. To reduce electricity consumption

The main objective of this project is to minimize unnecessary power usage. Traditional street lights remain ON for the entire night, even when there is no traffic. This system ensures that lights operate only when needed, thereby saving a significant amount of energy.

2. To automate street lighting system

This project focuses on eliminating manual control by using sensors and microcontrollers. The street lights automatically turn ON and OFF based on environmental conditions like darkness and movement, making the system fully automatic.

3. To increase efficiency using sensors

Sensors such as LDR and motion sensors are used to detect light intensity and movement. This improves the overall efficiency of the system by ensuring that lights are only used when required, reducing wastage and improving performance.

4. To minimize human intervention

By automating the entire process, the need for human involvement is reduced. There is no requirement for manual switching, which lowers maintenance effort and reduces chances of human error.

5. To develop a cost-effective smart solution

The system is designed using affordable components like Arduino, sensors, and LEDs. It provides a low-cost solution that can be easily implemented in both urban and rural areas, making it economically feasible.

Components Required

- **Arduino Uno**
  Main controller of the system that reads sensor data and controls the street lights.

- **LDR (Light Dependent Resistor)**
  Used to detect light intensity. It helps to identify day and night conditions.

- **IR Sensor / Ultrasonic Sensor**
  Used to detect movement of vehicles or people and turn ON the lights when needed.

- **LED / Street Light (Simulation)**
  Represents street lights in the project. It glows when the system activates.

- **Resistors**
  Used to control current and protect components from damage.

- **Jumper Wires**
  Used to connect different components in the circuit.

- **Breadboard**
  Used to build the circuit easily without soldering.

- **Power Supply**
  Provides power to the entire system.

Working Principle

The system works based on **light intensity and motion detection**:

- **Day Time:**
  LDR detects high light intensity → Street lights remain OFF
- **Night Time:**
  LDR detects low light → System activates
- **Vehicle/Person Detection:**
  Sensor detects movement → Lights turn ON
- **No Movement:**
  Lights remain DIM or OFF to save energy

Circuit Diagram

- LDR connected to analog pin of Arduino
- IR/Ultrasonic sensor connected to digital pin
- LED connected to output pin
- Power supply to Ardui

Algorithm

Start the system and initialize all components.

1. Read the value from the LDR sensor.
2. Check the light intensity:
   - If light intensity is high (daytime), turn OFF the LED.

3. If light intensity is low (nighttime), proceed to check the motion sensor.
4. Read the motion sensor value.
5. If motion is detected, turn ON the LED (street light).
6. If no motion is detected, keep the LED OFF or in DIM mode.
7. Repeat the process continuously for automatic operation.

Arduino Code

```
int ldrPin = A0;      // LDR connected to A0
int ledPin = 13;      // LED connected to pin 9
int ldrValue = 0;
void setup() {
  pinMode(ledPin, OUTPUT);
  Serial.begin(9600);
}
void loop() {
  ldrValue = analogRead(ldrPin);   // Read LDR value
  Serial.println(ldrValue);
  // Adjust threshold based on your environment
  if (ldrValue < 500) {
    digitalWrite(ledPin, LOW);
  } else {
    digitalWrite(ledPin, HIGH);
  }

  delay(500);
}
```

Applications

- Smart cities
- Highways
- Parking areas
- Residential streets
- Industrial areas

Future Enhancements

1. **Integration with Mobile App**
   The system can be connected to a mobile application, allowing users or authorities to monitor and control street lights remotely. This makes management easier and more efficient.

2. **Solar-Powered System**
   Solar panels can be added to power the street lights using renewable energy. This reduces dependency on electricity and makes the system more eco-friendly.

3. **Cloud Monitoring System**
   By connecting the system to the cloud, real-time data can be stored and monitored from anywhere. It helps in tracking performance and analyzing energy usage.

4. **Fault Detection System**
   An automatic fault detection feature can be added to identify issues like light failure or sensor malfunction. Alerts can be sent to maintenance teams for quick action.

5. **AI-Based Traffic Analysis**
Artificial Intelligence can be used to analyze traffic patterns and adjust lighting accordingly. This improves efficiency by providing light based on traffic density and usage patterns.

Conclusion

The Smart Street Light System using IoT is an efficient and cost-effective solution for modern cities. It helps in reducing energy consumption by operating lights only when required and minimizes human effort through automation. By using sensors and smart control, the system improves overall efficiency and reliability. This project plays an important role in developing smart and sustainable urban infrastructure.

References

- Arduino Official Website – https://www.arduino.cc
- IoT Tutorials and Project Guides (various online sources)
- Research papers on Smart Street Lighting Systems
- Electronics and Sensor Datasheets (LDR, IR Sensor, Ultrasonic Sensor)
- Basic IoT and Embedded Systems textbooks